Animal Matter
TO ME · SINCE 2010

PROJECT REPORT

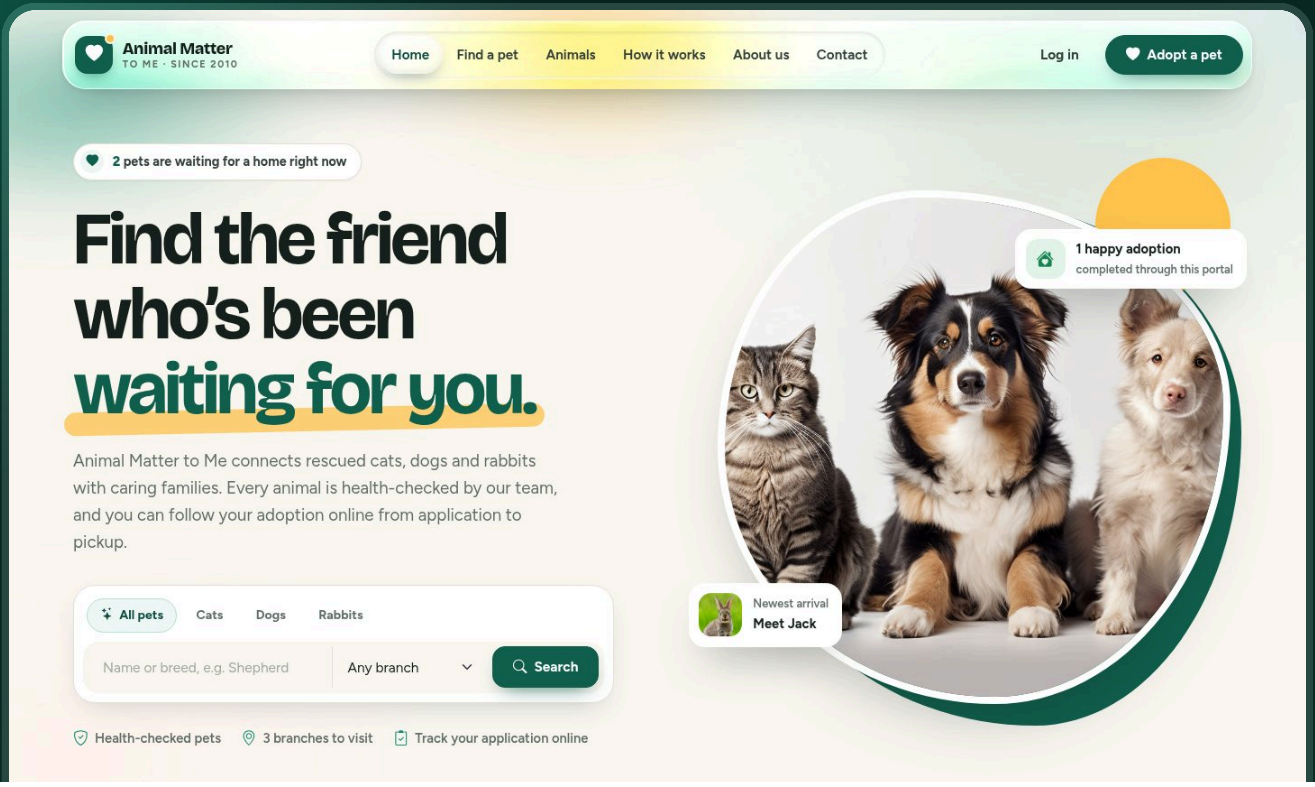
Pet Adoption Portal: features, working flow and system design

A web application that connects rescued cats, dogs and rabbits across several shelter branches with people who want to adopt them, from the first search to pickup and beyond.

PREPARED BY
Syed Hussain Ali Naqvi

DATE
October 2026

STACK
PHP 8 · MySQL · Bootstrap 5


Animal Matter
TO ME · SINCE 2010
Home
Find a pet
Animals
How it works
About us
Contact
Log in
Adopt a pet
2 pets are waiting for a home right now
1 happy adoption
completed through this portal
Newest arrival
Meet Jack
All pets
Cats
Dogs
Rabbits
Name or breed, e.g. Shepherd
Any branch
Search
Health-checked pets
3 branches to visit
Track your application online

Find the friend who's been waiting for you.

Animal Matter to Me connects rescued cats, dogs and rabbits with caring families. Every animal is health-checked by our team, and you can follow your adoption online from application to pickup.

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SECTION 1

Project overview

Animal Matter To Me (AMTM) is an online pet adoption portal for an animal welfare organisation that has rescued, treated and rehomed animals since 2010 and runs several shelter branches. The portal replaces phone calls and paper forms with one clear online process: people find a pet, apply to adopt it, follow their application step by step and collect the pet from the branch. The shelter team manages everything from a single admin portal.

3
user roles: visitor, adopter, shelter admin

40
page scripts across the public, adopter and admin areas

15
database tables (MySQL / MariaDB)

8
automatic emails sent at key moments

The problem it solves

- **Hard to discover pets.** Animals at different branches had no shared, searchable listing.
- **No transparency for adopters.** Applicants did not know whether their request had been seen, approved or declined, or why.
- **Scattered work for staff.** Pets, branches, applications, pickups and enquiries were tracked separately.
- **Contact ended at adoption.** There was no channel to support families after they took a pet home.

Goals

For adopters

- Search every available pet across all branches with useful filters.
- Apply online in a minute and see exactly where the application stands.
- Talk to the shelter directly once approved, and join adopter communities.

For the shelter

- Manage pets, animal types and branches in one place.
- Review, approve or decline requests with a recorded reason.
- Track pickups, adoption history and performance with live statistics.

Technology stack

Layer	Technology	Used for
Server	PHP 8.2 on Apache (XAMPP)	Page rendering, business rules, sessions
Database	MySQL / MariaDB with `mysqli` prepared statements	All application data (15 tables)
Interface	Bootstrap 5.3, Bootstrap Icons, custom design system (`assets/style.css`), Google Fonts	Responsive layout, components, “liquid glass” navigation
Interactivity	Vanilla JavaScript (Fetch API)	Live chat polling, instant filters, dialogs, previews
Charts	Chart.js 4.4.1	Admin dashboard trends
Email	PHPMailer over SMTP (e.g. Gmail app password)	Application, decision, reply and reset emails

SECTION 2

Users and roles

The portal has three kinds of user. Visitors can explore freely; creating a free account turns a visitor into an **adopter**; shelter staff sign in on the same login page and are routed to the **admin portal**. Every adopter and admin page checks the signed-in role on the server before doing anything.

Role	Can do	Highlights
Visitor not signed in	Browse and filter pets, view pet profiles and animal types, read about the shelter, send a contact message, sign up, log in, reset a forgotten password.	Live shelter statistics, newest arrivals, success stories, branch details and an FAQ on the homepage.
Adopter registered user	Everything a visitor can do, plus save favourites, apply to adopt, track and cancel applications, chat with the shelter after approval, take part in community groups, see replies to contact messages.	Personal dashboard, recommendations with reasons, pickup countdown, unread badges.
Admin shelter staff	Review requests (approve or reject with a reason), manage pickups and adoption history, manage pets, animal types and branches, answer messages, chat one-to-one with approved adopters, create and run community groups.	Analytics dashboard, notification bell, global pet search, performance metrics per branch and animal type.

How access is enforced

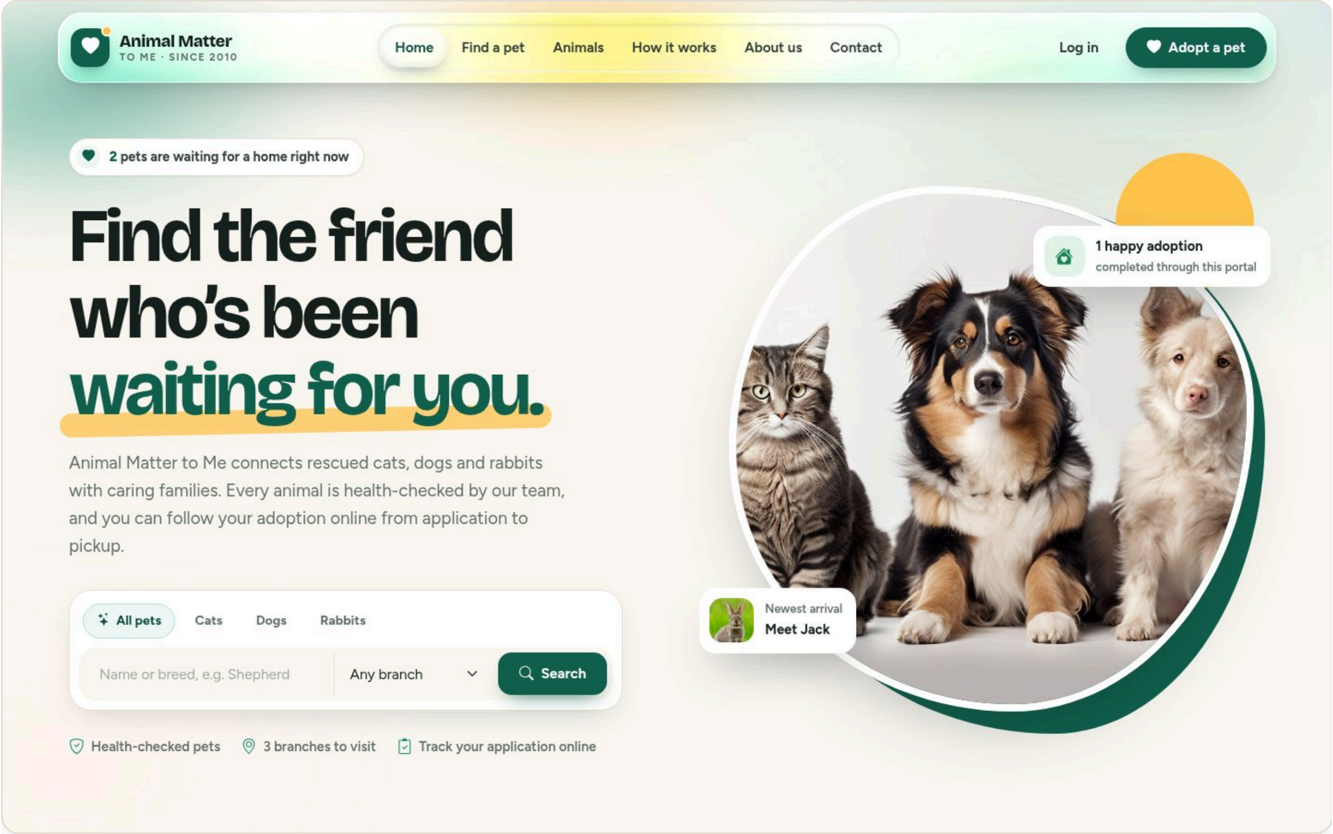
- One login form serves both adopters and staff. After a successful login the session stores the role and the user is sent to the right home page (or back to the page they were on, for example a pet they wanted to adopt).
- Every page in `adopter/` starts with `require_adopter()` and every page in `admin/` with `require_admin()`. Anyone else is redirected to the login page with a friendly message.
- Identity always comes from the server-side session, never from the URL, so one user cannot open another user's data by changing a link.

SECTION 3

Features

The portal is organised into three areas that share one design system: the **public website**, the **adopter area** and the **admin portal**. All screenshots below were taken from the running application.

3.1 Public website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3 branches to visit
Track your application online

Find the friend who's been waiting for you.

Animal Matter to Me connects rescued cats, dogs and rabbits with caring families. Every animal is health-checked by our team, and you can follow your adoption online from application to pickup.

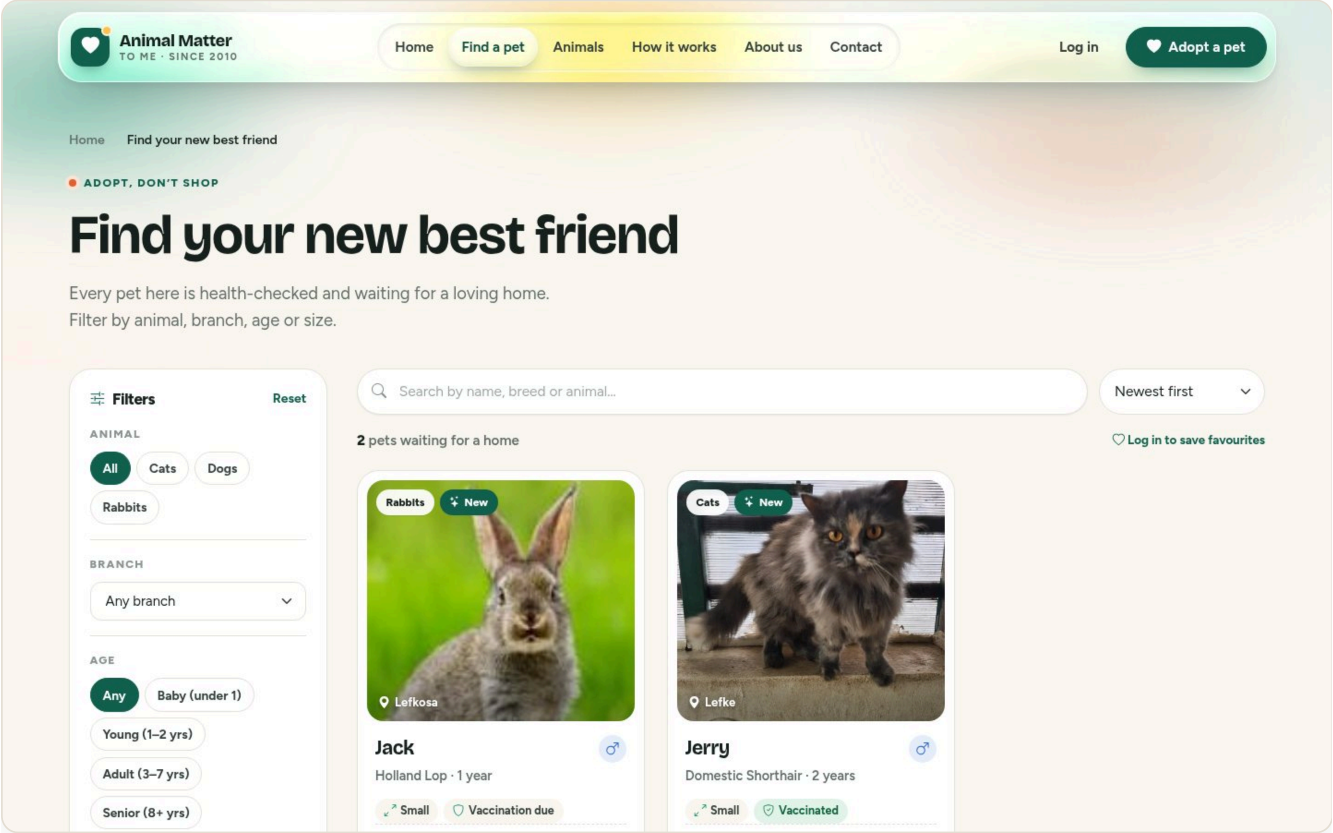
Figure 1. Homepage with the floating glass navigation, quick search by animal type, name or breed and branch, and live counters.

Homepage

- Hero search: animal type, name or breed, branch.
- **Shelter at a glance**: available now, adopted all-time and this year, average days from application to home, branches, newest arrival.
- Browse by animal, newest arrivals, "Recently found their family" (no adopter details shown), four-step explainer, services, branches with per-animal counts, FAQ.

Find a pet

- Filters: animal type, branch, age group (baby <1, young 1–2, adult 3–7, senior 8+), gender, size, vaccinated only, free-text search.
- Sorting (newest, name, youngest, oldest) and pages of 12.
- Cards show a *New* tag (listed in the last 7 days), days in care and how many people saved the pet.

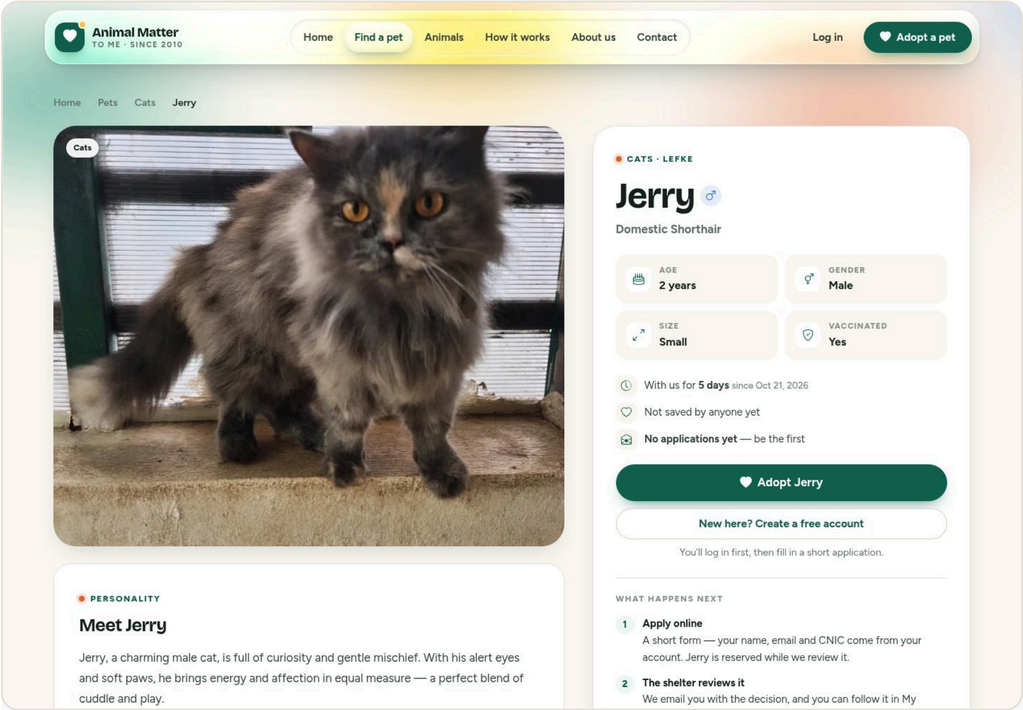

Animal Matter
TO ME · SINCE 2010
Home
Find a pet
Animals
How it works
About us
Contact
Log in
Adopt a pet
Home
Find your new best friend
ADOPT, DON'T SHOP

Find your new best friend

Every pet here is health-checked and waiting for a loving home.
Filter by animal, branch, age or size.

Filters
Reset
ANIMAL
All
Cats
Dogs
Rabbits
BRANCH
Any branch
AGE
Any
Baby (under 1)
Young (1–2 yrs)
Adult (3–7 yrs)
Senior (8+ yrs)
Search by name, breed or animal...
Newest first
2 pets waiting for a home
Log in to save favourites
Rabbits
New
Lefkosa
Jack
Holland Lop · 1 year
Small
Vaccination due
Cats
New
Lefke
Jerry
Domestic Shorthair · 2 years
Small
Vaccinated

Figure 2. Find a pet: filters on the left apply instantly; each card summarises age, gender, size, vaccination and time in care.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Pets
Cats
Jerry
Cats
CATS · LEFKE
Jerry
Domestic Shorthair

AGE: 2 years
GENDER: Male
SIZE: Small
VACCINATED: Yes

With us for **5 days** since Oct 21, 2026
Not saved by anyone yet
No applications yet — be the first

Adopt Jerry
New here? Create a free account

You'll log in first, then fill in a short application.

WHAT HAPPENS NEXT

1 **Apply online**
A short form — your name, email and CNIC come from your account. Jerry is reserved while we review it.

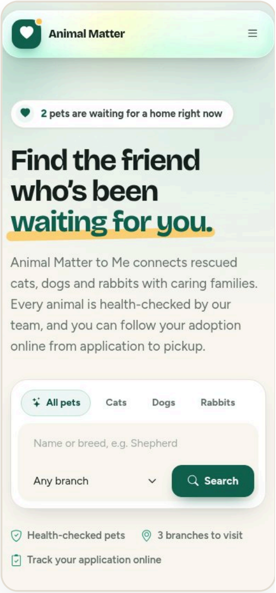
2 **The shelter reviews it**
We email you with the decision, and you can follow it in My

PERSONALITY

Meet Jerry

Jerry, a charming male cat, is full of curiosity and gentle mischief. With his alert eyes and soft paws, he brings energy and affection in equal measure — a perfect blend of cuddle and play.

Figure 3. Pet profile with facts, story, adoption checklist, "what happens next" steps and the branch card.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All pets
Cats
Dogs
Rabbits
Name or breed, e.g. Shepherd
Any branch
Search
Health-checked pets
3 branches to visit
Track your application online

Figure 4. Every page is responsive; on phones the navigation becomes a menu drawer.

Pet profile

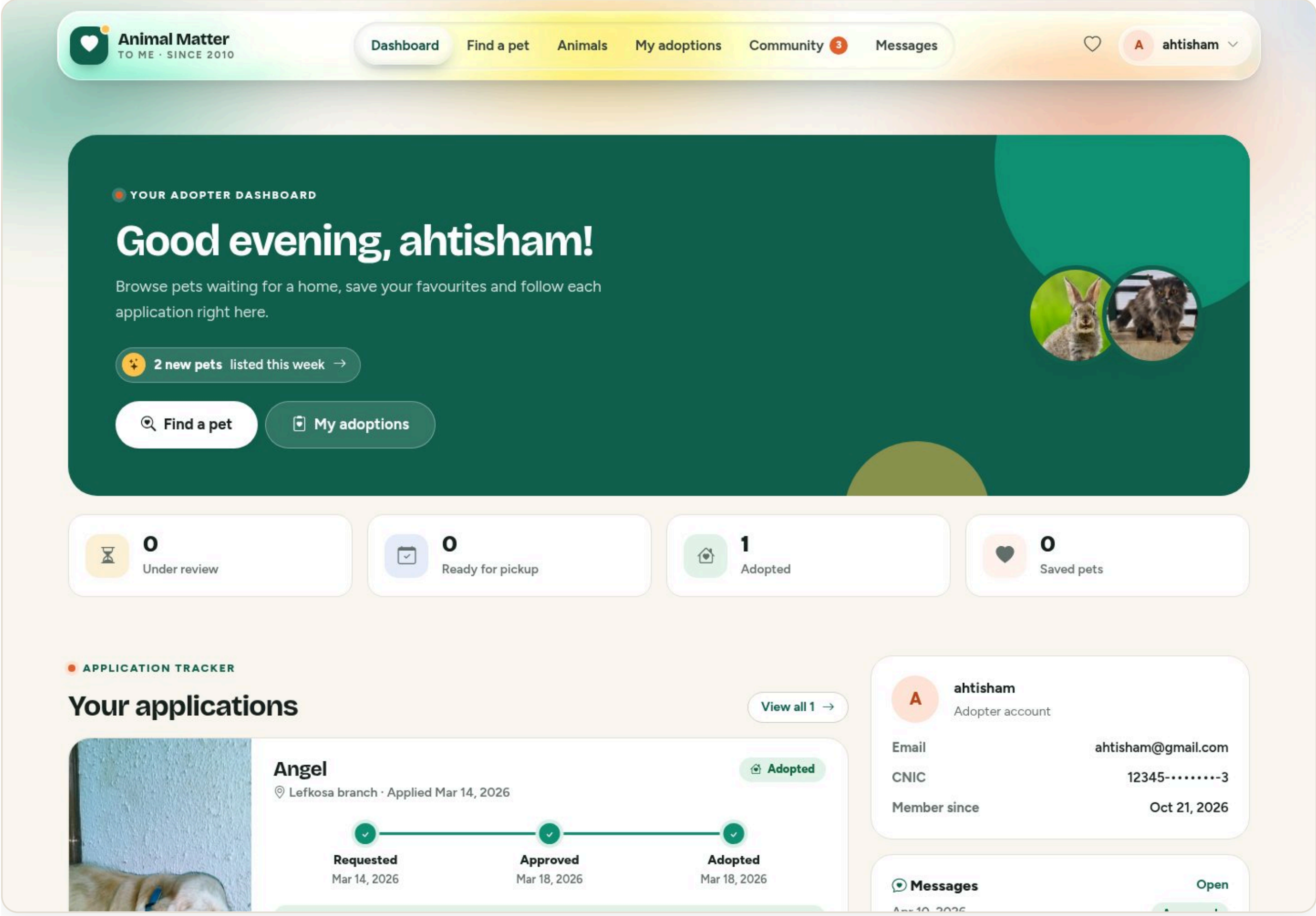
Each pet has its own page with photo, age, gender, size, vaccination status and story; how long it has been in care and how many adopters saved it; whether an application is already in progress; an **adoption checklist** (free account, 13-digit

CNIC, phone and address, pickup within 3 days of approval); the four steps that follow an application; the branch address and phone with a link to other pets at that branch; and similar pets.

Accounts

- **Sign up**: name, 13-digit CNIC (unique), email (unique across adopters and staff) and a password of at least 8 characters with a strength meter.
- **Log in**: one form for adopters and staff, with routing by role and “return to where you were”.
- **Forgot password**: a single-use reset link, valid for one hour, is emailed; the response never reveals whether an email is registered.
- **About us**, **animal types** with statistics, and a **contact** page with branch details and a message form.

3.2 Adopter area


Animal Matter
TO ME · SINCE 2010
Dashboard
Find a pet
Animals
My adoptions
Community 3
Messages
A ahtisham
YOUR ADOPTER DASHBOARD
2 new pets listed this week →
Find a pet
My adoptions
APPLICATION TRACKER
View all 1 →
Adopted
Open

Good evening, ahtisham!

Browse pets waiting for a home, save your favourites and follow each application right here.

0 Under review

0 Ready for pickup

1 Adopted

0 Saved pets

Your applications

Angel

Lefkosa branch · Applied Mar 14, 2026

Requested Mar 14, 2026

Approved Mar 18, 2026

Adopted Mar 18, 2026

ahtisham

Adopter account

Email	ahtisham@gmail.com
CNIC	12345-•••••••••-3
Member since	Oct 21, 2026

Messages

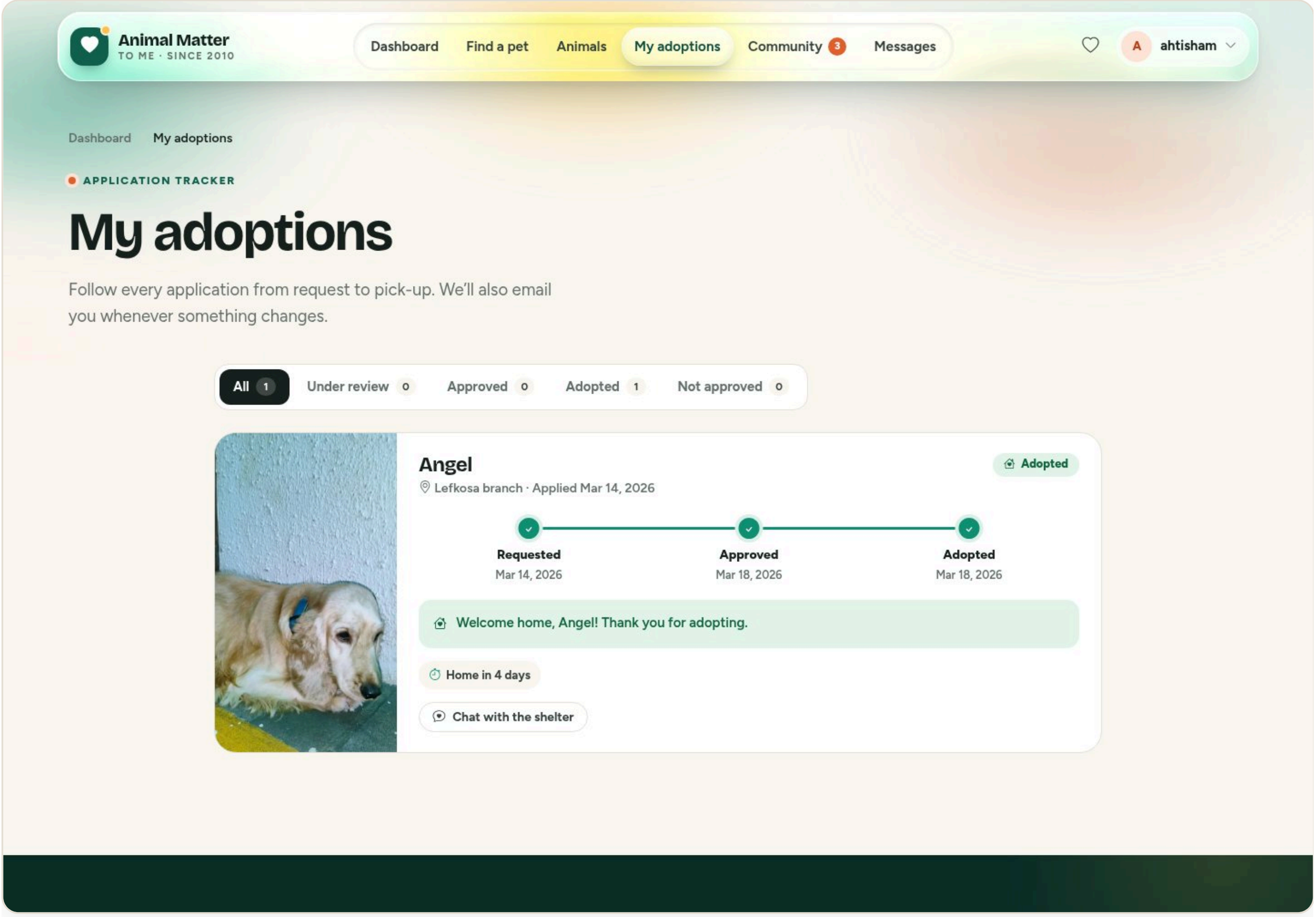
Figure 5. Adopter dashboard: greeting, counters, latest applications with their tracker, favourites and recommendations.

Dashboard and favourites

- Counters for applications under review, ready for pickup, adopted and saved pets.
- Heart button on every pet card to save favourites.
- Recommendations with a reason (“Because you saved a dog”, “New this week”).

Applying to adopt

- Form pre-filled from the account; adopter adds phone and address and confirms.
- The pet is reserved immediately; a database row lock stops two people applying for the same pet at once.
- A confirmation email is sent.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Dashboard
Find a pet
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Messages
A ahtisham
Dashboard
My adoptions
APPLICATION TRACKER

My adoptions

Follow every application from request to pick-up. We'll also email you whenever something changes.

All 1
Under review 0
Approved 0
Adopted 1
Not approved 0

Angel

Lefkosa branch · Applied Mar 14, 2026

Adopted

Requested Mar 14, 2026

Approved Mar 18, 2026

Adopted Mar 18, 2026

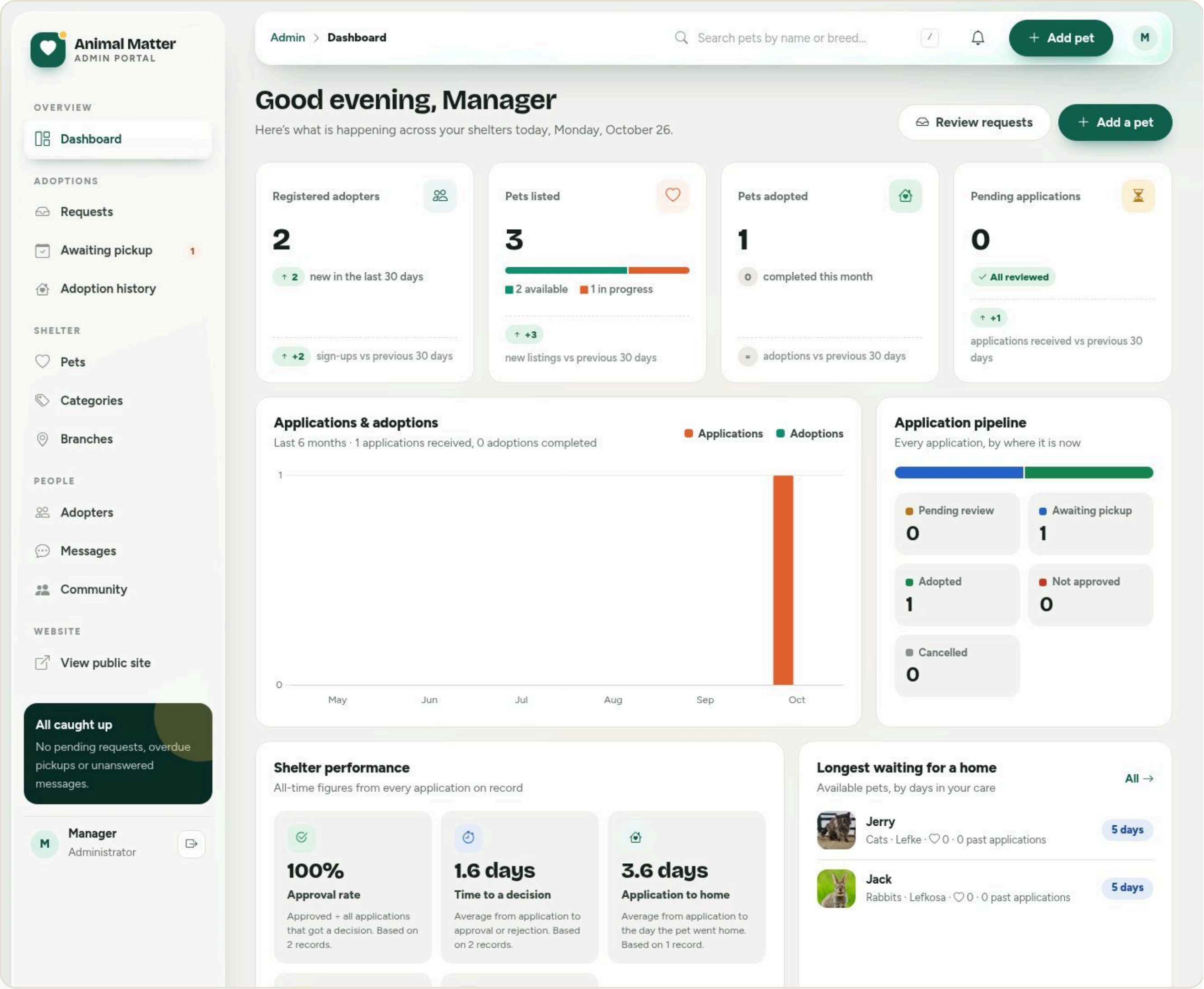
Welcome home, Angel! Thank you for adopting.

Home in 4 days
Chat with the shelter

Figure 6. My adoptions: tabs by status and a Requested → Approved → Adopted tracker for every application.

My adoptions lists every application with its stage, a “next step” line, the pickup deadline with a countdown (and an overdue warning), the branch phone, and the shelter’s reason if a request was not approved. Pending applications can be cancelled; approved and adopted ones link straight to the chat with the shelter.

3.3 Admin portal


Animal Matter
ADMIN PORTAL
OVERVIEW
Dashboard
ADOPTIONS
Requests
Awaiting pickup 1
Adoption history
SHELTER
Pets
Categories
Branches
PEOPLE
Adopters
Messages
Community
WEBSITE
View public site
Admin > Dashboard
Search pets by name or breed...
/
+ Add pet
M
Review requests
+ Add a pet
Applications
Adoptions
1
0
May
Jun
Jul
Aug
Sep
Oct
All →
5 days
5 days

All caught up

No pending requests, overdue pickups or unanswered messages.

Manager
Administrator

Good evening, Manager

Here's what is happening across your shelters today, Monday, October 26.

Registered adopters: **2** — ↑ 2 new in the last 30 days — ↑ +2 sign-ups vs previous 30 days

Pets listed: **3** — 2 available, 1 in progress — ↑ +3 new listings vs previous 30 days

Pets adopted: **1** — 0 completed this month — = adoptions vs previous 30 days

Pending applications: **0** — ✓ All reviewed — ↑ +1 applications received vs previous 30 days

Applications & adoptions

Last 6 months · 1 applications received, 0 adoptions completed

Application pipeline

Every application, by where it is now

Pending review 0
Awaiting pickup 1
Adopted 1
Not approved 0
Cancelled 0

Shelter performance

All-time figures from every application on record

100% Approval rate — Approved ÷ all applications that got a decision. Based on 2 records.

1.6 days Time to a decision — Average from application to approval or rejection. Based on 2 records.

3.6 days Application to home — Average from application to the day the pet went home. Based on 1 record.

Longest waiting for a home

Available pets, by days in your care

Jerry Cats · Lefke · ♡ 0 · 0 past applications — 5 days
Jack Rabbits · Lefkosa · ♡ 0 · 0 past applications — 5 days

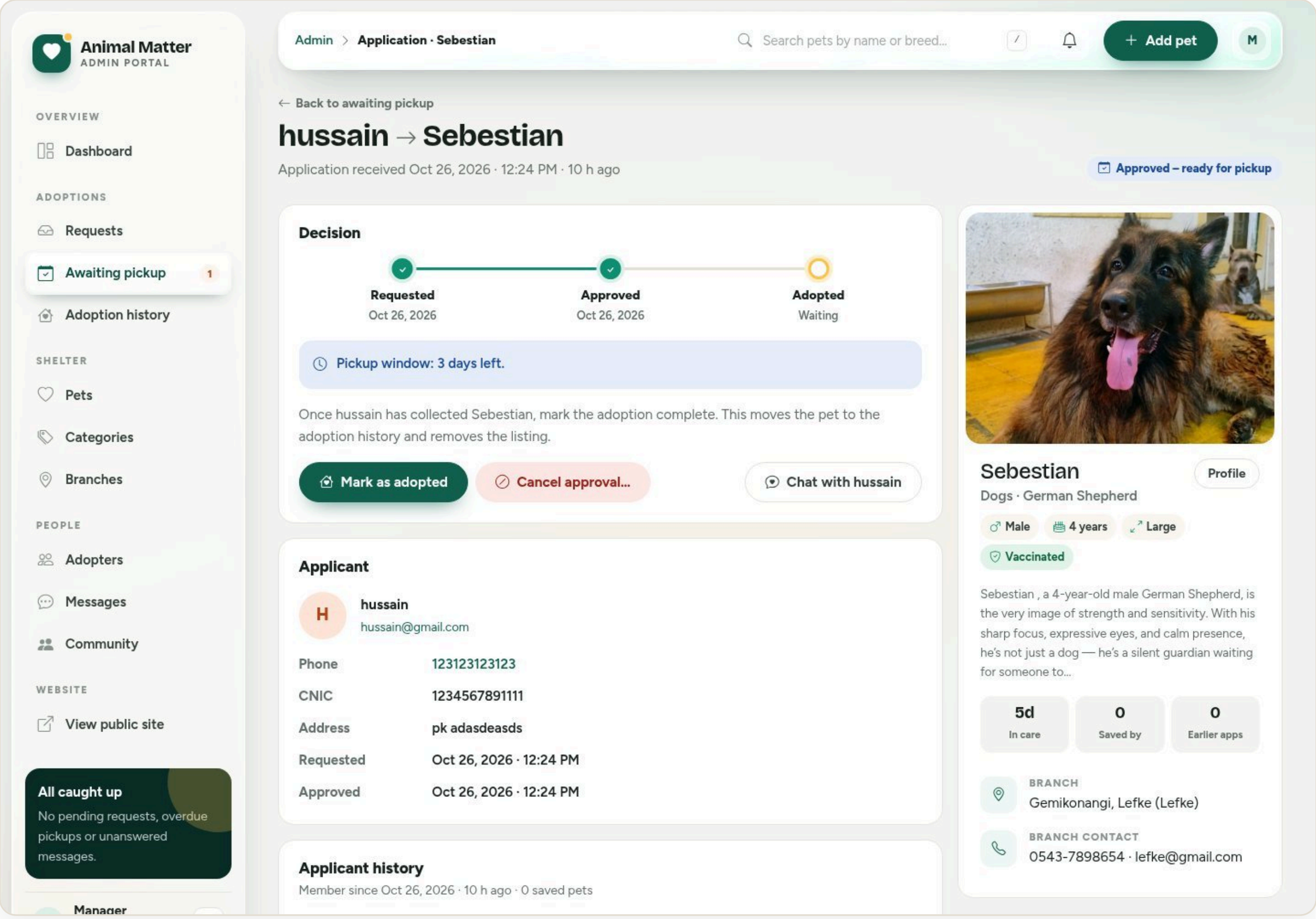
Figure 7. Admin dashboard: KPI cards with 30-day trends, applications and adoptions chart, pipeline, performance metrics and waiting lists.

Dashboard

- KPI cards (adopters, pets listed, adopted, pending) with change versus the previous 30 days.
- Six-month chart, application pipeline, **shelter performance** (approval rate, time to decision, application to home, on-time pickups, cancellation rate).
- Longest waiting for a home, most loved pets, per-animal and per-branch tables, recent activity, "needs your attention".

Everyday tools

- Notification bell for pending requests, overdue pickups, unread chats and community messages, unanswered contact messages and new adopters.
- Global pet search (press `/`) and quick "Add pet".
- Sidebar badges show what needs attention.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Adoption history
SHELTER
Pets
Categories
Branches
PEOPLE
Adopters
Messages
Community
WEBSITE
View public site
Manager
Admin › Application · Sebestian
Search pets by name or breed...
/
+ Add pet
M
← Back to awaiting pickup
Mark as adopted
Cancel approval...
Chat with hussain
Profile
H

All caught up

No pending requests, overdue pickups or unanswered messages.

hussain → Sebestian

Application received Oct 26, 2026 · 12:24 PM · 10 h ago

Approved – ready for pickup

Decision

Requested Oct 26, 2026 · Approved Oct 26, 2026 · Adopted Waiting

Pickup window: 3 days left.

Once hussain has collected Sebestian, mark the adoption complete. This moves the pet to the adoption history and removes the listing.

Applicant

hussain
hussain@gmail.com

Phone	123123123123
CNIC	1234567891111
Address	pk adasdeasds
Requested	Oct 26, 2026 · 12:24 PM
Approved	Oct 26, 2026 · 12:24 PM

Applicant history

Member since Oct 26, 2026 · 10 h ago · 0 saved pets

Sebestian

Dogs · German Shepherd

Male · 4 years · Large · Vaccinated

Sebestian , a 4-year-old male German Shepherd, is the very image of strength and sensitivity. With his sharp focus, expressive eyes, and calm presence, he's not just a dog — he's a silent guardian waiting for someone to...

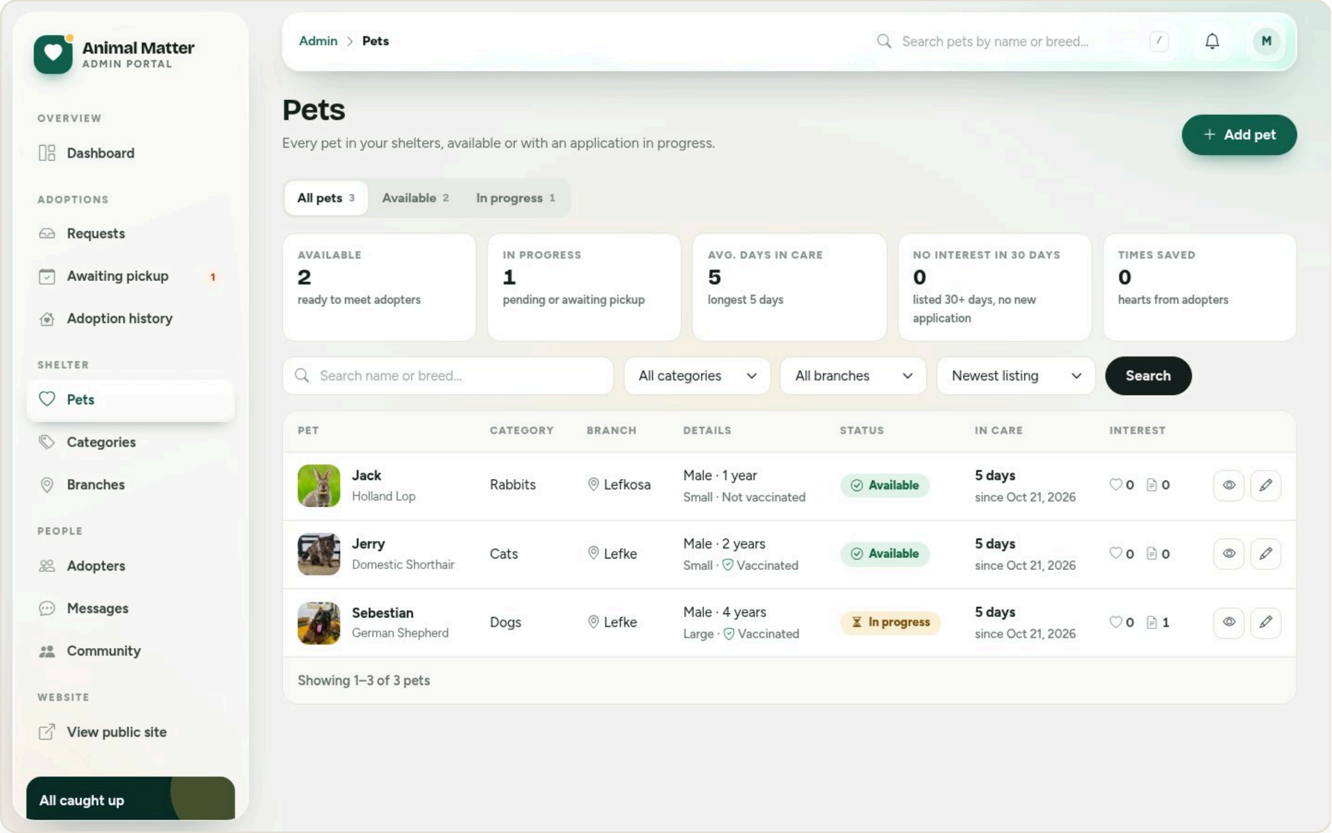
5d	0	0
In care	Saved by	Earlier apps

BRANCH
Gemikonangi, Lefke (Lefke)

BRANCH CONTACT
0543-7898654 · lefke@gmail.com

Figure 8. Application review: decision panel with pickup window, applicant details and history, and pet insights.

Module	What staff can do
Requests	Queue of pending applications with waiting time (amber after 2 days, red after 5) and applicant history; open a request to **approve** or **reject with a required reason**.
Awaiting pickup	Approved adopters with a pick-up-by deadline and countdown; **mark as adopted** after collection or **cancel the approval** with a reason; chat with the adopter.
Adoption history	Completed adoptions with days taken, plus an **Adopters** directory (joined, applications, last activity).
Pets	Searchable, sortable list (newest, longest in care, most saved, most applications, name) with days in care and interest; add and edit pets with photo upload; deletion blocked while an application is open.
Categories and branches	Create and edit animal types and shelter branches with performance figures; deletion blocked while they are in use.
Messages	Contact-form inbox (unanswered and answered, waiting time) with email replies; **Live chats**; **Community** groups.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All caught up
Admin > Pets
Search pets by name or breed...
/
M
+ Add pet
All pets 3
Available 2
In progress 1
Search name or breed...
All categories
All branches
Newest listing
Search

Pets

Every pet in your shelters, available or with an application in progress.

AVAILABLE 2 ready to meet adopters

IN PROGRESS 1 pending or awaiting pickup

AVG. DAYS IN CARE 5 longest 5 days

NO INTEREST IN 30 DAYS 0 listed 30+ days, no new application

TIMES SAVED 0 hearts from adopters

PET	CATEGORY	BRANCH	DETAILS	STATUS	IN CARE	INTEREST
Jack Holland Lop	Rabbits	Lefkosa	Male · 1 year Small · Not vaccinated	✓ Available	**5 days** since Oct 21, 2026	♡ 0 🗎 0
Jerry Domestic Shorthair	Cats	Lefke	Male · 2 years Small · ✓ Vaccinated	✓ Available	**5 days** since Oct 21, 2026	♡ 0 🗎 0
Sebastian German Shepherd	Dogs	Lefke	Male · 4 years Large · ✓ Vaccinated	⧗ In progress	**5 days** since Oct 21, 2026	♡ 0 🗎 1

Showing 1–3 of 3 pets

Figure 9. Pets list with summary strip, filters, sorting, status and interest columns.

3.4 Communication

The portal offers four ways for the shelter and adopters to stay in touch. All conversations update live in the browser without reloading the page.

Live one-to-one chat

- Opens automatically when an application is approved, and stays open after the adoption.
- Shows the approved or adopted pet at the top; Sent and Seen ticks; unread badges for both sides.
- Becomes read-only if the approval is cancelled.

Community groups

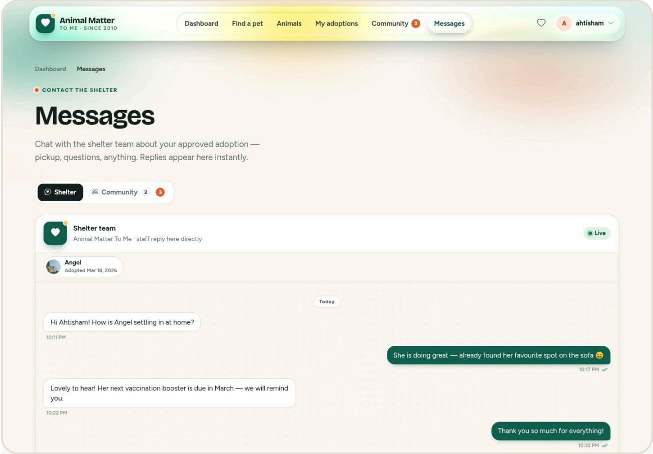
- Staff create groups (for example “Cat parents of Lefke”) and add adopters from a searchable list.
- Members chat with each other and the shelter team; names are shown, emails are not.
- Staff can rename, add or remove members and delete groups; members can leave.

Contact form

- Any visitor or adopter can write to the shelter.
- Staff reply from the inbox; the reply is emailed and appears on the adopter’s Messages page.

Email notifications

- Application received, approved (with pickup branch), not approved (with reason), approval cancelled, welcome home, contact reply, password reset, new chat message.
- Sent through PHPMailer only when SMTP is configured; otherwise skipped without breaking anything.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My adoptions
Community 3
Messages
A ahtisham
Dashboard
Messages
CONTACT THE SHELTER
Shelter
Community 2 3
Live
Today

Messages

Chat with the shelter team about your approved adoption — pickup, questions, anything. Replies appear here instantly.

Shelter team
Animal Matter To Me · staff reply here directly

Angel
Adopted Mar 18, 2026

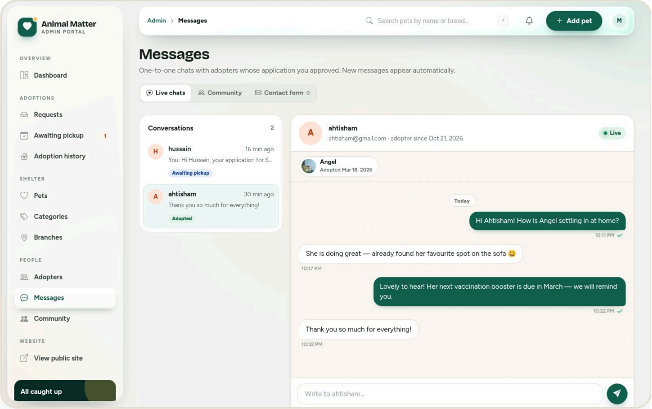
Hi Ahtisham! How is Angel settling in at home?
10:11 PM

She is doing great — already found her favourite spot on the sofa 😄
10:17 PM

Lovely to hear! Her next vaccination booster is due in March — we will remind you.
10:22 PM

Thank you so much for everything!
10:32 PM

Figure 10. Adopter side of the live chat with the shelter team.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PEOPLE
Adopters
Messages
Community
WEBSITE
View public site
All caught up
Admin > Messages
Search pets by name or breed...
Add pet
M
Live chats
Community
Contact form 0
Conversations 2
Awaiting pickup
Adopted
Live
Today
Write to ahtisham...

Messages

One-to-one chats with adopters whose application you approved. New messages appear automatically.

hussain 16 min ago
You: Hi Hussain, your application for S...

ahtisham 30 min ago
Thank you so much for everything!

ahtisham
ahtisham@gmail.com · adopter since Oct 21, 2026

Angel
Adopted Mar 18, 2026

Hi Ahtisham! How is Angel settling in at home?
10:11 PM

She is doing great — already found her favourite spot on the sofa 😄
10:17 PM

Lovely to hear! Her next vaccination booster is due in March — we will remind you.
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Thank you so much for everything!
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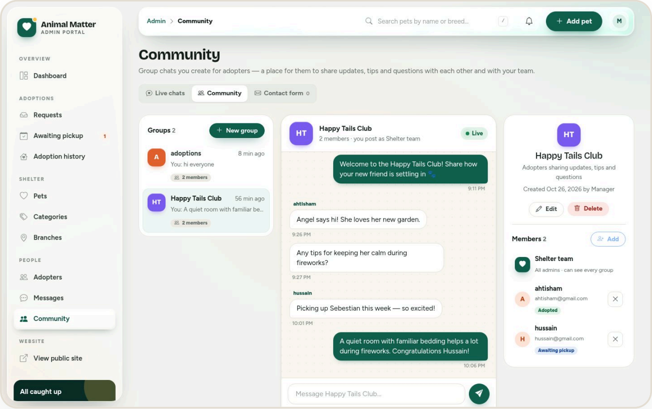
Figure 11. Admin side: conversation list with unread badges and the same chat.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Adopters
Messages
Community
WEBSITE
View public site
All caught up
Admin > Community
Search pets by name or breed...
Add pet
M
Live chats
Community
Contact form 0
Groups 2
New group
2 members
2 members
Live
Edit
Delete
Members 2
Add
Adopted
Awaiting pickup
Message Happy Tails Club...

Community

Group chats you create for adopters — a place for them to share updates, tips and questions with each other and with your team.

adoptions 8 min ago
You: hi everyone

Happy Tails Club 56 min ago
You: A quiet room with familiar be...

Happy Tails Club
2 members · you post as Shelter team

Welcome to the Happy Tails Club! Share how your new friend is settling in 🐾
9:11 PM

ahtisham
Angel says hi! She loves her new garden.
9:26 PM

Any tips for keeping her calm during fireworks?
9:27 PM

hussain
Picking up Sebastian this week — so excited!
10:01 PM

A quiet room with familiar bedding helps a lot during fireworks. Congratulations Hussain!
10:06 PM

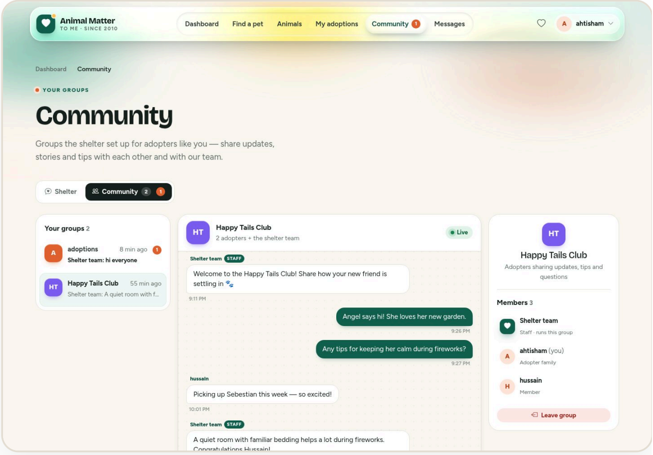
Happy Tails Club
Adopters sharing updates, tips and questions
Created Oct 26, 2026 by Manager

Shelter team
All admins · can see every group

ahtisham
ahtisham@gmail.com

hussain
hussain@gmail.com

Figure 12. Admin community view: groups, group chat and members panel.


Animal Matter
TO ME · SINCE 2010
Dashboard
Find a pet
Animals
My adoptions
Community 1
Messages
A ahtisham
Dashboard
Community
YOUR GROUPS
Shelter
Community 2 1
Your groups 2
Live
STAFF
STAFF
Members 3
Leave group

Community

Groups the shelter set up for adopters like you — share updates, stories and tips with each other and with our team.

adoptions 8 min ago
Shelter team: hi everyone

Happy Tails Club 55 min ago
Shelter team: A quiet room with f...

Happy Tails Club
2 adopters + the shelter team

Shelter team
Welcome to the Happy Tails Club! Share how your new friend is settling in 🐾
9:11 PM

Angel says hi! She loves her new garden.
9:26 PM

Any tips for keeping her calm during fireworks?
9:27 PM

hussain
Picking up Sebastian this week — so excited!
10:01 PM

Shelter team
A quiet room with familiar bedding helps a lot during fireworks. Congratulations Hussain!

Happy Tails Club
Adopters sharing updates, tips and questions

Shelter team
Staff · runs this group

ahtisham (you)
Adopter family

hussain
Member

Figure 13. An adopter in the same group, with names, a STAFF badge and system notes.

SECTION 4

Working flow

This section follows a pet and an adopter through the system: who does what, what the system records at each step and who is notified.

4.1 End-to-end journey

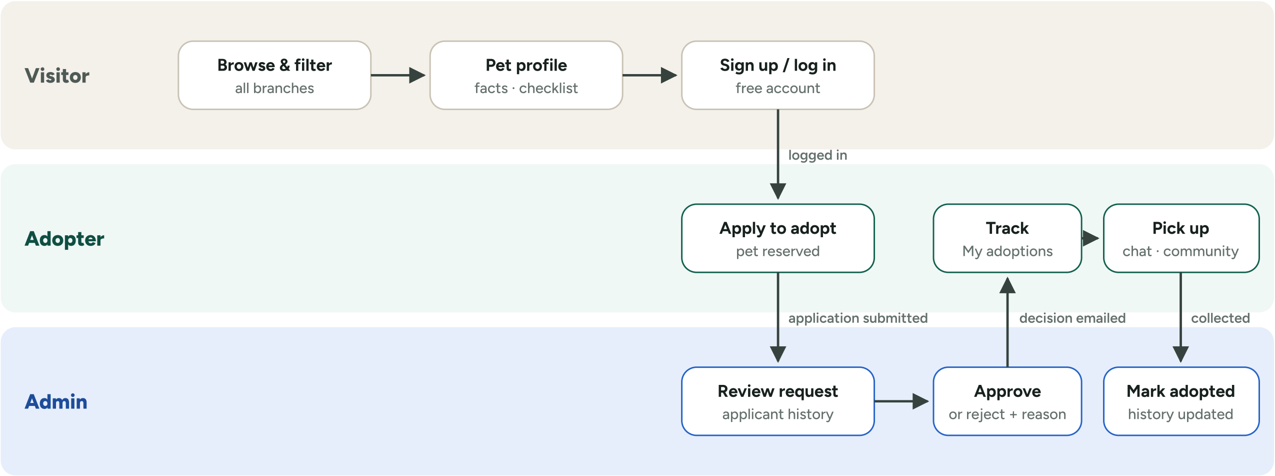

Visitor
Browse & filter
all branches
Pet profile
facts · checklist
Sign up / log in
free account
logged in
Adopter
Apply to adopt
pet reserved
Track
My adoptions
Pick up
chat · community
application submitted
decision emailed
collected
Admin
Review request
applicant history
Approve
or reject + reason
Mark adopted
history updated

Figure 14. The main path through the system, by role. Arrows crossing lanes are hand-overs between people, each backed by an email and a status change.

4.2 Adoption lifecycle

Every pet and application moves through a small set of states. Only the actions shown can change a state, and each action is recorded in the database and, where relevant, emailed to the adopter.

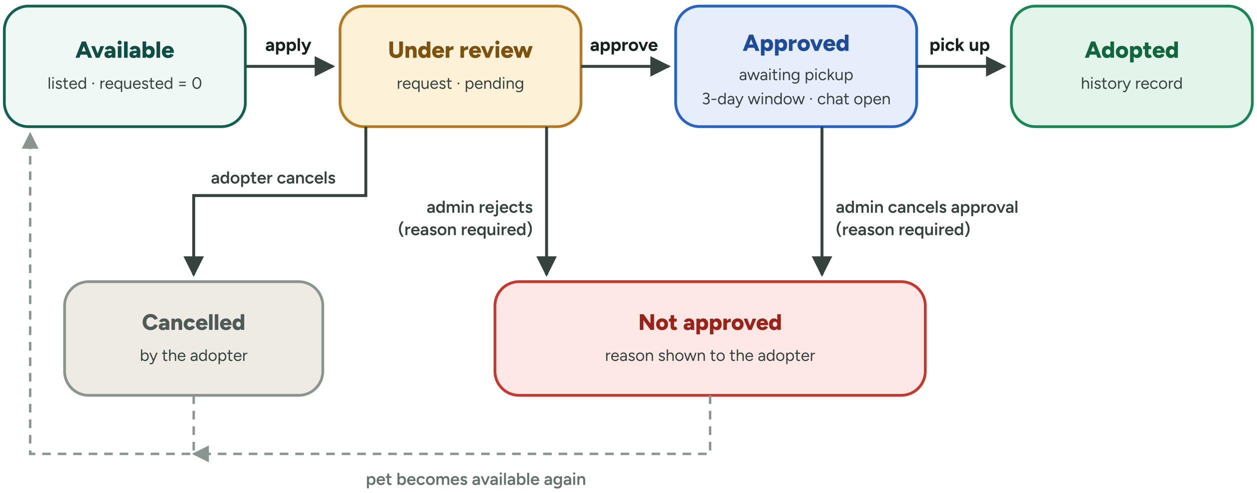

Available
listed · requested = 0
apply
Under review
request · pending
approve
Approved
awaiting pickup
3-day window · chat open
pick up
Adopted
history record
adopter cancels
admin rejects
(reason required)
admin cancels approval
(reason required)
Cancelled
by the adopter
Not approved
reason shown to the adopter
pet becomes available again

Figure 15. Adoption lifecycle. Solid arrows are actions by the adopter or the shelter; dashed arrows show the pet returning to the listings.

Action	By	What the system does	Adopter is told
Apply	Adopter	Checks the pet is still free (row lock), creates a `requests` row with status pending, sets the pet to requested so it leaves the listings.	"We received your application" email
Cancel	Adopter	Only while pending: status becomes cancelled, the pet is listed again.	Shown in My adoptions
Reject	Admin	Reason required: status rejected with the reason saved, the pet is listed again.	Email with the reason; reason in tracker
Approve	Admin	Moves the application to `accepted` with the approval time; the 3-day pickup window starts; one-to-one chat unlocks.	Email with branch address; countdown
Cancel approval	Admin	Reason required: removes the approval, records a rejected request with the reason, lists the pet again; chat becomes read-only.	Email with the reason
Mark adopted	Admin	Creates an `adopted` history record (pet, branch and all three dates), removes the listing and clears favourites for it, in one transaction.	"Welcome home" email; tracker complete

4.3 Accounts and sign-in

1. **Sign up.** The visitor enters name, CNIC, email and password. The server validates every field, checks the CNIC and email are unique, stores a bcrypt hash of the password, signs the user in and opens the adopter dashboard.
2. **Log in.** The same form checks adopters first, then staff. On success the session ID is renewed, the role is stored and the user is sent to their dashboard or to the page they came from.
3. **Forgot password.** A random token is generated, only its SHA-256 hash is stored, and a link valid for one hour is emailed. The same message is shown whether or not the email exists.
4. **Reset.** A valid, unexpired token lets the user choose a new password (8+ characters). The token is deleted so the link cannot be reused.

4.4 Live one-to-one chat

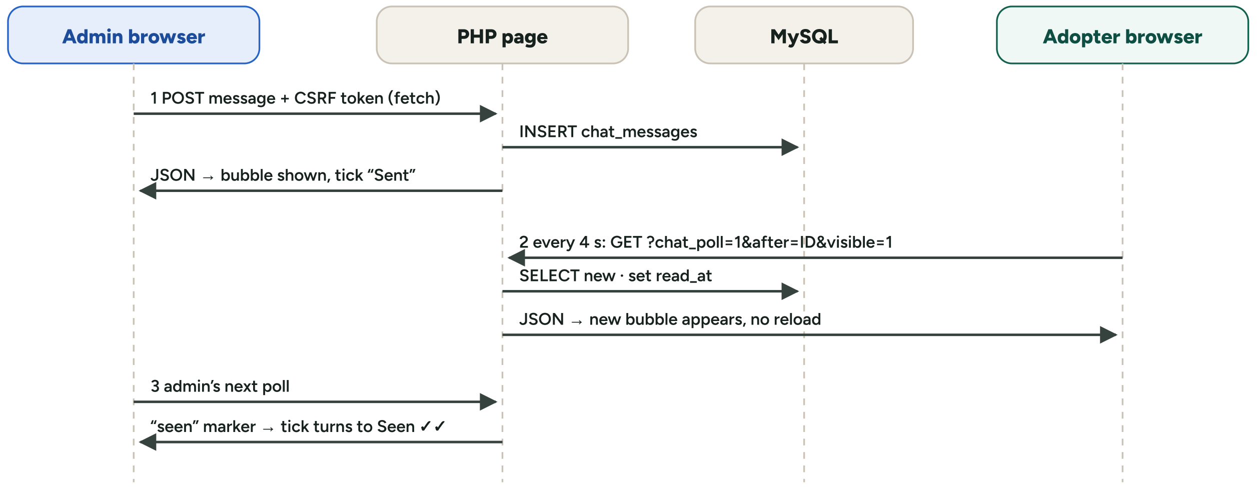

Admin browser
PHP page
MySQL
Adopter browser
1 POST message + CSRF token (fetch)
INSERT chat_messages
JSON → bubble shown, tick "Sent"
2 every 4 s: GET ?chat_poll=1&after=ID&visible=1
SELECT new · set read_at
JSON → new bubble appears, no reload
3 admin's next poll
"seen" marker → tick turns to Seen ✓✓

Figure 16. How a chat message travels. The same pattern runs in both directions; group chats use it without the Seen ticks.

- **When it opens.** The chat is available once the adopter has an approved application (or a completed adoption). Before that, the adopter sees a note explaining it will unlock after approval.
- **Sending.** Enter sends, Shift + Enter adds a new line, messages up to 1,000 characters. If JavaScript is off, the form still works with a normal page reload.
- **Unread and email.** Unread counts appear in the navigation and the admin notification bell. The adopter is emailed about the first unread message from the shelter, not every message.

4.5 Community groups

1. **Create.** An admin opens Community → New group, enters a name (3–60 characters) and optional description, and ticks adopters to add. The group starts with notes such as “Manager created the group” and “Ali joined the group”.
2. **Chat.** Members and staff post in the group; each person’s name appears above their messages, staff messages carry a STAFF badge, and new messages appear live with unread badges in the navigation.
3. **Manage.** Admins can add or remove members, rename the group or delete it with all messages. Members can leave. Every change leaves a note in the chat.
4. **Access.** Only members (and staff) can open a group or receive its messages; removed members lose access immediately. New members can read earlier messages but start with nothing unread.

4.6 Contact form

A visitor or adopter sends a message (5–500 characters) from the contact page. It appears in the admin inbox under *Unanswered*, with how long it has been waiting. When staff reply, the answer is saved and emailed; signed-in adopters also see the reply under their message history.

SECTION 5

System architecture

The portal is a classic server-rendered PHP application. Each page is a PHP script; all shared logic lives in one core file, `dbconnect.php`, so every page follows the same rules for database access, security and layout.

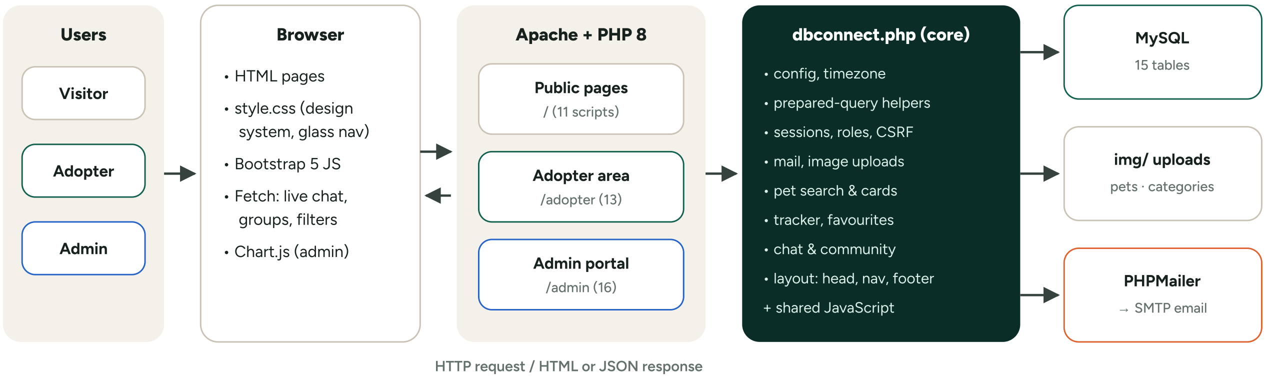

Users
Visitor
Adopter
Admin
Browser
• HTML pages
• style.css (design system, glass nav)
• Bootstrap 5 JS
• Fetch: live chat, groups, filters
• Chart.js (admin)
Apache + PHP 8
Public pages
/ (11 scripts)
Adopter area
/adopter (13)
Admin portal
/admin (16)
dbconnect.php (core)
• config, timezone
• prepared-query helpers
• sessions, roles, CSRF
• mail, image uploads
• pet search & cards
• tracker, favourites
• chat & community
• layout: head, nav, footer
+ shared JavaScript
MySQL
15 tables
img/ uploads
pets · categories
PHPMailer
→ SMTP email
HTTP request / HTML or JSON response

Figure 17. Components. Pages in all three areas call the shared core for data, security and layout.

Life of a request

Every page script follows the same pattern, which keeps behaviour predictable and makes the code easy to extend:

1. **Bootstrap.** `require_once dbconnect.php` loads configuration, connects to MySQL (timezone aligned with PHP), starts the session and, for any POST, verifies the CSRF token.
2. **Authorise.** Adopter and admin pages call `require_adopter()` or `require_admin()` and get the signed-in user's ID from the session.
3. **Handle actions.** Form posts are validated, written with prepared statements (inside a transaction when several tables change), followed by a short flash message and a redirect, so refreshing never repeats an action. Live chat requests are answered with JSON and stop here.
4. **Render.** The page sets `$pageTitle` and `$activeNav`, includes its area's `header.php` (head, navigation, notifications), prints its content with shared helpers such as `render_pet_card()` and `render_chat()`, and finishes with `page_end()` (footer, confirmation dialog and shared JavaScript).

Code layout

```
pet_adoption.sql            database schema + sample data
pet_adoption_portal/
├── dbconnect.php            shared core (config, DB, auth, CSRF, mail, uploads, pets, chat, community, layout)
├── header.php               public layout (glass navigation, mobile drawer)
├── index.php · pets.php · petInfo.php · categories.php · sub_categories.php
├── aboutUs.php · contactUs.php · signup.php · login.php · forgetPassword.php · resetPassword.php
├── adopter/                 header.php, home, pets, categories, categoryPets, petInfo, adoptForm,
│                            requested (+ accepted, adopted tabs), removeRequest, contactUs (chat, community)
├── admin/                   header.php (sidebar, top bar, bell), home (dashboard), requests, requestResponse,
│                            accepted, adopted, pets, petView, petEdit, categories, categoryEdit, categoryPets,
│                            locations, locationEdit, contact (inbox, live chats, community), contactResponse
├── assets/style.css         design system: tokens, components, responsive rules, liquid glass
├── img/                     site photos, pets/ and category/ uploads (random file names)
└── phpmailer/               email library
```

Design system. One stylesheet defines colours, type, spacing and every component. The stylesheet link carries a version number taken from the file’s modification time, so browsers always load the latest styles after an update.

SECTION 6

Database design

The database (`pet_adoption`) has 15 InnoDB tables using UTF-8 (utf8mb4). The diagram shows how they connect; lines are links through ID columns.

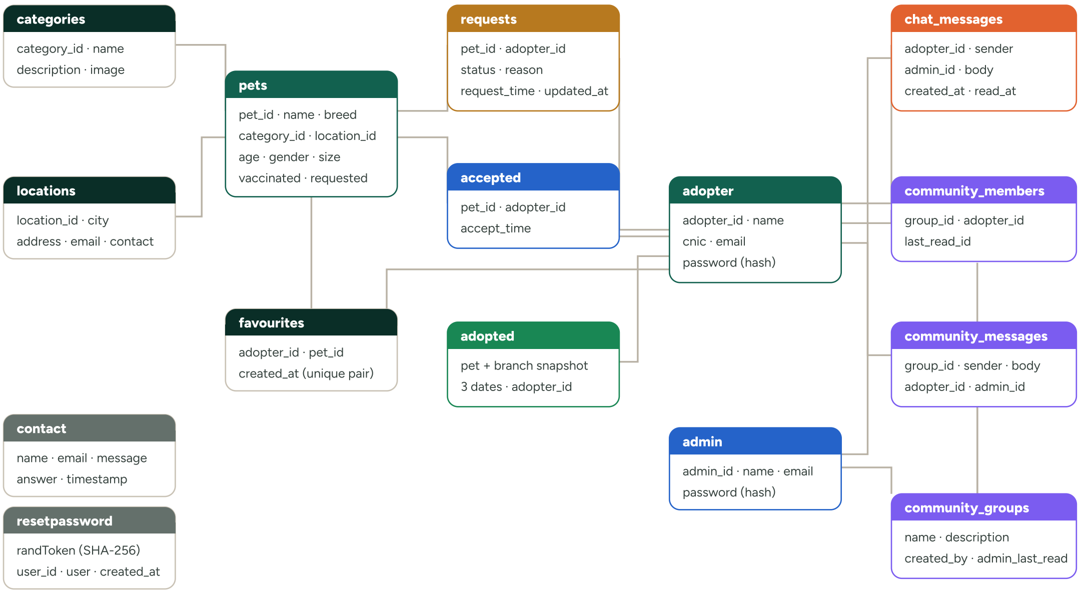

categories
category_id · name
description · image
requests
pet_id · adopter_id
status · reason
request_time · updated_at
chat_messages
adopter_id · sender
admin_id · body
created_at · read_at
pets
pet_id · name · breed
category_id · location_id
age · gender · size
vaccinated · requested
locations
location_id · city
address · email · contact
accepted
pet_id · adopter_id
accept_time
adopter
adopter_id · name
cnic · email
password (hash)
community_members
group_id · adopter_id
last_read_id
favourites
adopter_id · pet_id
created_at (unique pair)
adopted
pet + branch snapshot
3 dates · adopter_id
community_messages
group_id · sender · body
adopter_id · admin_id
contact
name · email · message
answer · timestamp
admin
admin_id · name · email
password (hash)
resetpassword
randToken (SHA-256)
user_id · user · created_at
community_groups
name · description
created_by · admin_last_read

Figure 18. Entity relationships. Green: people and pets; amber, blue, dark green: the three adoption stages; orange: one-to-one chat; purple: community.

Table	Purpose and key columns
pets	Every listed animal: name, breed, age, gender, size, vaccinated, story, image, category and branch, `requested` flag (0 = available, 1 = application in progress), `created_at` .
categories, **locations**	Animal types (with image and description) and shelter branches (city, address, email, phone).
adopter, **admin**	Accounts. Passwords are stored only as bcrypt hashes; adopters also have a unique CNIC and a sign-up date.
requests	Applications with contact details, `status` (pending, rejected, cancelled), `reason` , request and update times.
accepted	Approved applications waiting for pickup, with `accept_time` (start of the 3-day window).
adopted	Permanent history: a snapshot of the pet and branch plus request, approval and adoption dates, so records survive after the listing is removed.
favourites	Pets each adopter saved (one row per adopter and pet).
contact	Contact-form messages and the shelter's answer.
resetpassword	Hashed password-reset tokens with creation time (valid for one hour).
chat_messages	One-to-one chat: one conversation per adopter, `sender` (admin or adopter), body, sent and read times.
community_groups, **community_members**, **community_messages**	Groups, who is in them (with each member's last read message) and the group conversation, including system notes.

SECTION 7

Security and data protection

Security is built into the shared core, so every page gets the same protection without extra code.

Threat	Protection in the portal
SQL injection	Every query uses prepared statements with placeholders through `db_query()`, `db_all()`, `db_one()` and `db_run()`; no user value is ever placed in SQL text.
Stolen or weak passwords	Passwords are hashed with bcrypt (`password_hash`) and checked with `password_verify`; minimum 8 characters. Older plain-text passwords are upgraded to a hash at the next login.
Unauthorised access	Session-based roles checked on every adopter and admin page; the session ID is renewed at login; user IDs come from the session, never from the URL.
Cross-site request forgery	A per-session token is required on every POST and checked automatically before any page logic runs.
Cross-site scripting	All output is escaped with `htmlspecialchars`; chat and group messages are inserted as plain text, so HTML in a message is displayed, not run.
Malicious uploads	Images are checked by their real content (JPG, PNG, WEBP or GIF), limited to 5 MB and saved under random names.
Password reset abuse	Random 64-character tokens, stored only as SHA-256 hashes, valid for one hour and deleted after use; the same message is shown whether or not an email exists.
Race conditions and data loss	Multi-step changes run in transactions; applying locks the pet row so two people cannot reserve the same pet; categories, branches and pets cannot be deleted while they are in use.
Privacy	Public pages never show adopter names or details; community members see each other's names but not emails; only staff see applicant contact details.

Before going live. Change the sample admin login (`manager@gmail.com` / `manager`), set a MySQL password, keep real SMTP credentials out of version control, and serve the site over HTTPS.

SECTION 8

Installation and running

Requirements

XAMPP (Apache, PHP 8.2 or newer, MySQL / MariaDB) on Windows, macOS or Linux, and a modern browser. Internet access is used for Bootstrap, icons, fonts and Chart.js from CDNs, and for sending email.

Steps

1. Copy the `pet_adoption_portal` folder into XAMPP's `htdocs` folder (on Linux, `/opt/lampp/htdocs`). The folder name must stay `pet_adoption_portal`.
2. Start Apache and MySQL from the XAMPP control panel (on Linux: `sudo /opt/lampp/lampp start`).
3. Create a database named `pet_adoption` (utf8mb4) in phpMyAdmin and import `pet_adoption.sql`.
4. Make sure the web server can write to `img/` so admins can upload photos.
5. Open `http://localhost/pet_adoption_portal/`.

Sample logins

Role	Email	Password	Opens
Admin	`manager@gmail.com`	`manager`	Admin dashboard
Adopter	`ahtisham@gmail.com`	`ahtisham`	Adopter dashboard

Configuration

All settings are at the top of `dbconnect.php`:

- **Database**: host, user, password and database name (XAMPP defaults: `localhost`, `root`, no password).
- **Email**: `SMTP_USER`, `SMTP_PASS` and `SMTP_FROM`. For Gmail, use an App Password. While these are empty the portal works normally and simply skips sending.
- **Timezone**: `APP_TIMEZONE` (default `Asia/Karachi`) keeps PHP and MySQL on the same clock, so “2 min ago”, chat times and pickup deadlines are correct.

SECTION 9

Testing and quality

The portal was checked at several levels while it was being built. All checks ran against the real application and database, and test data was removed afterwards.

Check	What was verified
Syntax	`php -l` on every PHP file after each change.
Page crawl	Every public, adopter and admin page loaded as a visitor, an adopter and an admin: all return HTTP 200 with no PHP warnings, notices or errors, including with an empty database (no division by zero, "—" shown instead).
Adoption flow	Apply → reject with reason → apply again → approve → mark adopted, end to end over HTTP, with the database checked after each step; cancelling twice, applying twice for the same pet and plain GET requests to the cancel action are refused.
Admin management	Adding and editing pets, categories and branches with image upload (fake images and missing fields rejected), guarded deletes, message replies.
Security	Requests without a CSRF token are rejected; HTML and script in messages are shown as text; pages and chat endpoints refuse users without the right role or membership.
Live chat and community	Real-browser tests (headless Chrome): send with Enter without reloading, replies appear by polling, Sent turns to Seen, names shown once per run of messages, staff badge, no ticks in groups.
Layout	Screenshots of every key page at desktop (1440 px) and phone (390 px) widths, reviewed and fixed where content overflowed.

SECTION 10

Limitations and future work

Current limitations

- **Near-real-time, not instant.** Chats refresh every 4 seconds by polling. This is simple and reliable on shared hosting, but a WebSocket server would scale better for many users.
- **Shared shelter inbox.** All staff share one chat identity (“Shelter team”) and one read state; there is no assignment of conversations to individual staff.
- **Text-only conversations.** Chat and community messages cannot include photos or files yet.
- **Email depends on SMTP.** Without SMTP settings, notifications are only visible inside the portal.
- **Adoption history stores names as text.** Per-branch and per-animal statistics match on the stored city and category name; renaming a branch or category separates older records.

Possible next steps

Adopters

- Photo sharing in community groups
- Book a pickup time slot online
- Browser push notifications
- Urdu / multi-language interface

Shelter staff

- Assign chats and requests to staff members
- Medical and vaccination records per pet
- Home-visit scheduling before approval
- Export reports to CSV or PDF

Platform

- WebSockets for instant messaging
- Audit log of admin actions
- Two-factor login for staff
- Automated test suite and CI

Conclusion

Animal Matter To Me turns a manual, phone-based adoption process into a clear online journey. Adopters can find pets across all branches, apply in minutes and always know where their application stands. Staff manage pets, branches, decisions and pickups from one dashboard with live statistics. The live chat and community features keep the shelter connected with families after adoption. The code is organised around one shared core with consistent security rules, which makes the portal straightforward to run on XAMPP and to extend.

APPENDIX A

Page inventory

Page	Area	Purpose
`index.php`	Public	Homepage: search, live statistics, animals, newest arrivals, happy tails, branches, FAQ
`pets.php`	Public	Find a pet: filters, sorting, pagination
`petInfo.php`	Public	Pet profile, checklist, branch, similar pets
`categories.php`, `sub_categories.php`	Public	Animal types and pets of one type, with statistics
`aboutUs.php`, `contactUs.php`	Public	About the shelter; branches and contact form
`signup.php`, `login.php`	Public	Create an adopter account; sign in for adopters and staff
`forgetPassword.php`, `resetPassword.php`	Public	Request and use a password reset link
`adopter/home.php`	Adopter	Dashboard: counters, applications, favourites, recommendations
`adopter/pets.php`, `categories.php`, `categoryPets.php`, `petInfo.php`	Adopter	Browsing with favourites and personalised actions
`adopter/adoptForm.php`	Adopter	Application form
`adopter/requested.php` (+ `accepted.php`, `adopted.php`)	Adopter	My adoptions tracker with status tabs
`adopter/removeRequest.php`	Adopter	Cancels a pending application (POST only)
`adopter/contactUs.php`	Adopter	Messages: live chat with the shelter, community groups, contact form and history
`admin/home.php`	Admin	Analytics dashboard
`admin/requests.php`, `requestResponse.php`	Admin	Request queue; review, approve, reject, mark adopted, cancel approval
`admin/accepted.php`, `adopted.php`	Admin	Awaiting pickup; adoption history and adopters directory
`admin/pets.php`, `petView.php`, `petEdit.php`	Admin	Pet list, details, add and edit with photos
`admin/categories.php`, `categoryEdit.php`, `categoryPets.php`	Admin	Animal types and their pets
`admin/locations.php`, `locationEdit.php`	Admin	Shelter branches
`admin/contact.php`, `contactResponse.php`	Admin	Inbox and replies, live chats, community groups
`*/logout.php`	Both	Ends the session